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Development Economics & Econometrics Cheatsheet

A compact reference for common models, estimators, and diagnostics.

Quick Reference

Topic	Formula / Guidance
Core Statistics	• **Mean/Variance:** Mean: $\bar{y} = (1/n)\sum y_i$; Variance: $s^2 = (1/(n-1))\sum(y_i-\bar{y})^2$ • **Covariance/Correlation:** $Cov(y,x) = (1/(n-1))\sum(y_i-\bar{y})(x_i-\bar{x})$; $\rho = Cov(y,x)/(s_y s_x)$ • **t-statistic:** $t = (\hat{\beta} - \beta_0)/SE(\hat{\beta})$ • **Confidence Interval:** $\hat{\beta} \pm t_{\alpha/2,df} \times SE(\hat{\beta})$
OLS & Inference	• **OLS:** $\hat{\beta} = (X'X)^{-1}X'y$; requires Exogeneity ($E[u\|X]=0$), No perfect multicollinearity, SLR assumptions. • **Heteroskedasticity-robust SE:** Use HC1/HC2/HC3; in Stata: regress y x ... , vce(robust). • **Goodness-of-fit:** $R^2 = 1 - (SSR/TSS)$; Adjusted R^2 adjusts for k. • **Omitted Variable Bias:** Bias $\approx \delta \times Corr(X_1, Z) \times$ effect of Z on y; include Z or use IV.
Causal Designs	• **IV / 2SLS:** Stage1: $\hat{X} = \Pi Z + ...$; Stage2: $y = \beta\hat{X} + ...$ Assumptions: Relevance, Exclusion, Independence. • **RDD:** Sharp: D=1{running var ≥ c}; local linear on both sides; bandwidth selection critical.

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	• **Diff-in-Diff:** $y_{it} = \alpha + \beta(D_t \times T_i) + \gamma_i + \delta_t + \varepsilon_{it}$; parallel trends key. • **Synthetic Control:** Weighted donor pool to match pre-treatment path; placebo tests are essential.
Panels & FE	• **FE:** $y_{it} = \alpha_i + \beta x_{it} + \varepsilon_{it}$; demeans by unit. • **RE:** Assumes $\alpha_i \perp x_{it}$; Hausman test compares FE vs RE. • **Clustered SEs:** Cluster where shocks occur (e.g., village, worker, firm).
Limited Dependent Vars	• **Logit/Probit:** $\Pr(y=1\|x) = F(x\beta)$; marginal effects depend on x; report APEs or at means. • **Poisson/NegBin:** Counts; Poisson with robust SE consistent under misspecification (QMLE).
Diagnostics	• **Multicollinearity:** Check VIF; center variables; drop redundant terms. • **Endogeneity:** Use IV/design; Durbin-Wu-Hausman test. • **Heteroskedasticity:** White/Breusch-Pagan; use robust/clustered SE. • **Autocorrelation:** Durbin-Watson; Newey-West for time series.

Tip: Always report your identifying assumptions and robust checks alongside estimates.

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